

MH1300 FOUNDATIONS OF MATHEMATICS

Tutorial 12 Solutions

Ex. 8.1.11. Let $A = \{3, 4, 5\}$ and $B = \{4, 5, 6\}$ and let S be the “divides” relation. That is,

$$\text{for all } (x, y) \in A \times B, \quad x S y \iff x|y.$$

State explicitly which ordered pairs are in S and S^{-1} ?

SOLUTION . By inspection,

$$S = \{(3, 6), (4, 4), (5, 5)\}.$$

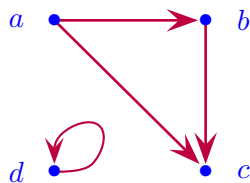
Therefore, interchanging the first and second coordinate of each ordered pair in S ,

$$S^{-1} = \{(4, 4), (5, 5), (6, 3)\}.$$

□

Ex. 8.1.14. Define a relation S on $B = \{a, b, c, d\}$ by $S = \{(a, b), (a, c), (b, c), (d, d)\}$. Draw the directed graphs of the relation S .

SOLUTION . Directed graph:



□

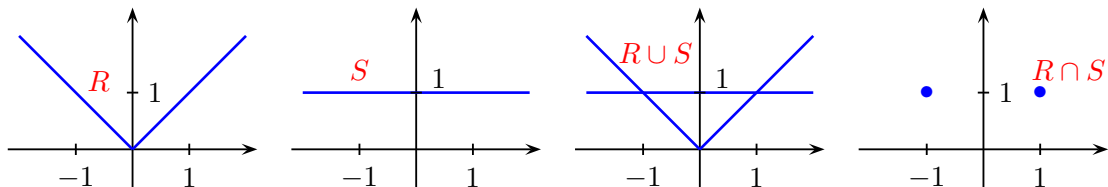
Ex. 8.1.22. Define relations R and S from $\mathbb{R}$ to $\mathbb{R}$ as follows:

$$R = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = |x|\} \quad \text{and}$$

$$S = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = 1\}.$$

Graph R , S , $R \cup S$, and $R \cap S$ in the Cartesian plane.

SOLUTION .



□

Ex. 8.2.17, 21. Determine whether the given relation is reflexive, symmetric, transitive, or none of these. Justify your answers.

17. Recall that a prime number is an integer that is greater than 1 and has no positive integer divisors other than 1 and itself. (In particular, 1 is not prime.) A relation P is defined on $\mathbb{Z}$ as follows:

$$\text{For all } m, n \in \mathbb{Z}, \quad m P n \iff \exists \text{ a prime number } p \text{ such that } p|m \text{ and } p|n.$$

21. Let $X = \{a, b, c\}$ and $\mathcal{P}(X)$ be the power set of X . A relation $\mathbf{L}$ is defined on $\mathcal{P}(X)$ as follows:

$$\text{For all } A, B \in \mathcal{P}(X), \quad A \mathbf{L} B \iff |A| < |B|.$$

($|A| < |B|$ means the number of elements in A is less than the number of elements in B .)

SOLUTION . 17.

a. P is not reflexive.

Consider the integer 1. By Theorem 4.3.2, we know that the only divisors of 1 are 1 and -1 . Since prime numbers are greater than 1, no prime numbers are divisors of 1. Therefore $\nexists$ prime number p such that $p|1$ and $p|1$. Therefore $1 \not P 1$.

b. P is symmetric.

Suppose m and n are integers such that $m P n$. Then for some prime number p , we have $p|m$ and $p|n$.

Therefore, for the same prime number p , $p|n$ and $p|m$. Therefore, $n P m$.

c. P is not transitive. Consider the integers 2, 6, 9. Then 2 is a prime number and $2|2$ and $2|6$, hence $2 P 6$. Also, 3 is a prime number and $3|6$ and $3|9$, hence $6 P 9$. However, $2 \not P 9$ since 2 and 9 do not have any common prime factor.

21.

a. $\mathbf{L}$ is not reflexive.

Consider the set $A = \{a\}$ which is a subset of X . Then $|A| = 1$ and $1 \not< 1$. Therefore, $|A| \not< |A|$. Therefore $(A, A) \notin \mathbf{L}$.

b. **L** is not symmetric.

Consider the subsets $A = \emptyset$ and $B = \{a\}$ of X . Then $|A| = 0$ and $|B| = 1$. Since $0 < 1$, we have $(A, B) \in \mathbf{L}$. Since $1 \not< 0$, we know that $(B, A) \notin \mathbf{L}$.

c. **L** is transitive.

Suppose A , B and C are subsets of X such that $A \mathbf{L} B$ and $B \mathbf{L} C$. Then we know that $|A| < |B|$ and $|B| < |C|$. By the transitivity of $<$, we have $|A| < |C|$. Hence $A \mathbf{L} C$. □

Ex. 8.2.40–42. Suppose R and S are relations on a set A .

40. If R and S are reflexive, is $R \cup S$ reflexive? Why?

41. If R and S are symmetric, is $R \cup S$ symmetric? Why?

42. If R and S are transitive, is $R \cup S$ transitive? Why?

SOLUTION . 40. Yes. Let a be an element in A . Since R is reflexive, we know that $(a, a) \in R$. Hence, by the definition of $\cup$, we have $(a, a) \in R \cup S$.

41. Yes. Suppose we are given $a, b \in A$ such that $(a, b) \in R \cup S$. Then, either $(a, b) \in R$ or $(a, b) \in S$.

Case 1: $(a, b) \in R$. Since R is symmetric, we have $(b, a) \in R$. Hence, by definition of $\cup$, $(b, a) \in R \cup S$.

Case 2: $(a, b) \in S$. Since S is symmetric, we have $(b, a) \in S$. Hence, by definition of $\cup$, $(b, a) \in R \cup S$.

42. No, $R \cup S$ need not be transitive.

Counterexample:

Consider $A = \{a, b, c\}$, and $R = \{(a, b)\}$ and $S = \{(b, c)\}$. Then $R \cup S = \{(a, b), (b, c)\}$. Clearly R and S are transitive (true by default since there is nothing to check).

We note that $(a, b) \in R \cup S$ and $(b, c) \in R \cup S$ but $(a, c) \notin R \cup S$. □

Ex. 8.3.15. Determine which of the following congruence relations are true and which are false.

a. $17 \equiv 2 \pmod{5}$

b. $4 \equiv -5 \pmod{7}$

c. $-2 \equiv -8 \pmod{3}$

d. $-6 \equiv 22 \pmod{2}$.

SOLUTION . a. True since $17 - 2 = 15 = 5 \cdot 3$ and hence $5|15$.

b. False since $4 - (-5) = 9$ and $7 \nmid 9$.

c. True since $-2 - (-8) = 6 = 3 \cdot 2$ and hence $3|6$.

d. True since $-6 - 22 = -28 = 2 \cdot (-14)$ and hence $2|(-28)$. □

Ex. 8.3.32. Let A be the set of all straight lines in the Cartesian plane. Define a relation $\parallel$ on A as follows:

$$\text{For all } l_1 \text{ and } l_2 \text{ in } A, \quad l_1 \parallel l_2 \iff l_1 \text{ is parallel to } l_2.$$

Prove that $\parallel$ is an equivalence relation. Describe the distinct equivalence classes of the relation.

SOLUTION . 1. $\parallel$ is reflexive.

Let line $l \in A$ be given. Then clearly $l \parallel l$ since every l is parallel to l .

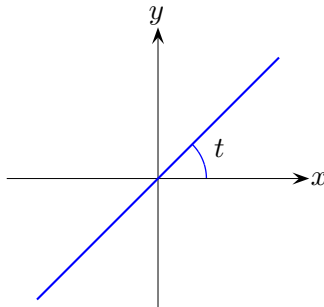
2. $\parallel$ is symmetric. Suppose l_1 and l_2 are two straight lines in A such that l_1 is parallel to l_2 . Then l_2 is parallel to l_1 and so $l_2 \parallel l_1$.

3. $\parallel$ is transitive.

Suppose three straight lines l_1, l_2 , and l_3 are given where $l_1 \parallel l_2$ and $l_2 \parallel l_3$. In other words, l_1 is parallel to l_2 , and l_2 is parallel to l_3 . Therefore, l_1 is parallel to l_3 , and so $l_1 \parallel l_3$.

Since $\parallel$ is reflexive, symmetric, and transitive, we gather that $\parallel$ is an equivalence relation.

Next, we describe the distinct equivalence classes. There are two ways to do this. First, we use the fact that any straight line is parallel to a line passing through the origin. We may consider the angle t that the line makes with the positive x -axis.



Any line through the origin makes an angle t with the positive x -axis, where $0 \leq t < \pi$. Also, for any two lines l_1 and l_2 making angles t_1 and t_2 , respectively, with the positive x -axis, where $0 \leq t_1, t_2 < \pi$ and $t_1 \neq t_2$, we know that $l_1 \nparallel l_2$. Therefore, the distinct equivalence classes of $\parallel$ are the lines through the origin making angles t with the positive x -axis for each t in $0 \leq t < \pi$. (If you have studied polar curves before, we may represent the distinct equivalence classes of $\parallel$ by the polar equations

$$\theta = t \text{ for each } t \text{ in } 0 \leq t < \pi.)$$

Another way to describe the distinct equivalence classes is to note that the equation of a line is given by,

$$y = mx + C, \quad -\infty < m < \infty,$$

where m is the gradient of the line and C is the y -intercept, or

$$x = D,$$

when the line is vertical and D is the x -intercept. Since any two lines are parallel if and only if their gradients are equal, we may choose x -intercepts and y -intercepts as zero for each equivalence class. Therefore, the distinct equivalence classes are

$$[y = mx], \quad -\infty < m < \infty, \quad \text{and} \quad [x = 0].$$

□

Ex. 8.3.33. Let A be the set of points in the rectangle with x and y coordinates between 0 and 1. That is,

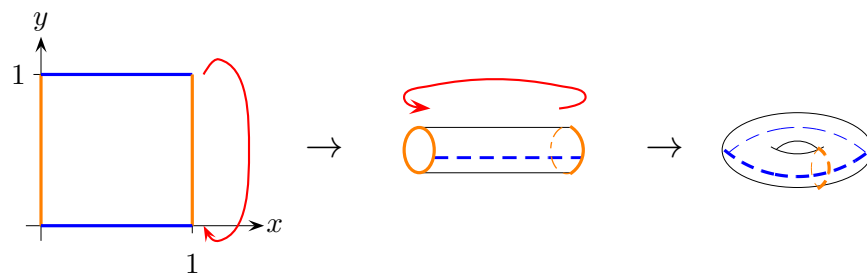
$$A = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1\}.$$

Define a relation R on A as follows: For all (x_1, y_1) and (x_2, y_2) in A ,

$$\begin{aligned} (x_1, y_1) R (x_2, y_2) \iff & (x_1, y_1) = (x_2, y_2); \text{ or} \\ & x_1 = 0 \text{ and } x_2 = 1 \text{ and } y_1 = y_2; \text{ or} \\ & x_1 = 1 \text{ and } x_2 = 0 \text{ and } y_1 = y_2; \text{ or} \\ & y_1 = 0 \text{ and } y_2 = 1 \text{ and } x_1 = x_2; \text{ or} \\ & y_1 = 1 \text{ and } y_2 = 0 \text{ and } x_1 = x_2; \text{ or} \\ & (x_1, y_1), (x_2, y_2) \in \{(0, 0), (0, 1), (1, 0), (1, 1)\}. \end{aligned}$$

In other words, all points along the top edge of the rectangle are related to the points along the bottom edge directly beneath them, and all points directly opposite each other along the left and right edges are related to each other. The points in the interior of the rectangle are not related to anything other than themselves. Then R is an equivalence relation on A . Imagine gluing together all the points that are in the same equivalence class. Describe the resulting figure.

SOLUTION .



The distinct equivalence classes can be identified with the points on a geometric figure, called a torus, that has the shape of the surface of a doughnut (or donut). Each point in the interior of the rectangle $\{(x, y) \mid 0 < x < 1 \text{ and } 0 < y < 1\}$ is only equivalent to itself. Each point on the top edge of the rectangle is in the same equivalence class as the point vertically below it on the bottom edge of the rectangle (so we can imagine identifying these points by gluing them together – this gives us a cylinder), and each point on the left edge of the rectangle is in the same equivalence class as the point horizontally across from it on the right edge of the rectangle (so we can also imagine identifying these points by gluing them together – this brings the two ends of the cylinder together to produce a torus).

View a computer rendering at:

http://www.youtube.com/watch?gl=SG&v=OH5_h-RB0T8&hl=en-GB.

□

Ex. 8.3.46. Let R be a relation on a set A and suppose R is a symmetric and transitive relation. Prove the following:

If for every x in A there is a y in A such that $x R y$, then R is an equivalence relation.

SOLUTION . Suppose R is a relation on a set A , R is symmetric and transitive, and for every element $x \in A$, there exists a $y \in A$ such that $x R y$. Let x be any element in A . [Want to show $x R x$.]

According to the hypothesis, there exists a $y \in A$ such that $x R y$. By symmetry, we have $y R x$. Since $x R y$ and $y R x$, by transitivity, we have $x R x$. This shows that R is reflexive.

Since R is reflexive, symmetric, and transitive, we conclude that R is an equivalence relation. □